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SUPERCEDING
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FEDERAL SPECIFICATION

HANDLES, WOOD, ACME THREADED AND TAPERED ENDS

This Federal Specification has been approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.2 Scope. This specification covers straight wood handles with tapered or threaded ends, for use with brooms, brushes, squeegees, paint rollers, etc.

1.3 Classification. The handles furnished under this specification shall be of the following types, sizes and lengths as specified (see 6.2):

Type I - ACME threaded metal end.
Type II - Tapered end.

Size 1 - 7/8 inch (22 mm) in diameter.
Size 2 - 15/16 inch (24 mm) in diameter.
Size 3 - 1-1/8 inch (29 mm) in diameter.

Lengths - 4 ft (1.22 m), 4-1/2 ft (1.37 m), 5 ft (1.52 m), and 8 ft (2.44 m) (or as otherwise specified).

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 The documents referenced in this specification shall be the issues in effect on the date of the invitation for bids or request for proposal unless otherwise specified. These form a part of this specification to the extent specified. In the event that there is a conflict between this specification and a document referenced herein, this specification shall take precedence.

Beneficial comments, recommendations, additions, deletions, clarifications, etc., and any information which might improve this document should be sent to: General Services Administration, Engineering Group (7FXEE), 819 Taylor Street, Fort Worth, TX 76102.

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FSC 7920

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Federal Standards:

FED-STD-376 - Preferred Metric Units for General use by the Federal Government.

Copies of Federal Standards are available from the General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

Military Standard:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Copies of Military Standards may be obtained from the Naval Publications and Forms Center, Documents Order Desk, Bldg. 4D, 700 Robins Avenue, Philadelphia, PA 19111-5094.

ASTM:

D 4444 - Standard Test Methods for Use and Calibration of Hand-Held Moisture Meters

E 29 - Standard Practice For Using Digits in Test Data to Determine Conformance With Specifications.

Applications for copies of ASTM (American Society for Testing and Materials) standards should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.

3. REQUIREMENTS

3.1 Wood. The type I and type II handles shall be constructed of close-grained and straight grained ash, beech, birch, cherry, elm, hickory, ramin (Gonystylus spp.), pecan, maple, red oak, or white oak. Type II handles may also be constructed of Douglas fir or southern yellow pine. The handles shall not be fabricated from woods with a cross grain or spiral grain sloping more than 1 inch in 15 inches (2.54 cm in 38.10 cm). The wood shall be dried, seasoned, and prepared prior to fabrication. The finished handle shall have a moisture content no more than 12% when tested as specified in 4.2.2.2.

3.2 Construction. The diameter of the handle shall be as specified (see 6.2) and shall be uniform throughout it's length. The handle shall be straight and shall not have a crook, warp, or bow that causes an eccentricity of more than 3/8 inch (0.95 cm) when the handle is rolled on a flat surface (4.2.1.1). The handle ends opposite the attachment end shall be rounded. The attachment end for the type I handles shall have a threaded ferrule as specified in 3.2.1. The attachment end for the type II shall be tapered as specified in 3.2.2.

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3.2.1 Type I end. The type I handle shall have a metal ferrule with ACME threads permanently attached to one end. The ferrule shall be composed of aluminum or corrosion resistant steel. The threads shall conform to the dimensions of figure 1. The end of the ferrule shall be rounded or beveled as indicated. The attachment of the ferrule to the handle shall be sufficient to withstand the attachment test (see 4.2.2.1).

3.2.2 Type II end. The type II handles shall be tapered on one end, the dimensions shall be as specified by figure 2. The taper shall be straight, uniform, and with a round cross-section.

3.3 Size. Size 1 handles shall be 7/8 inch (22 mm) in diameter. Size 2 handles shall be 15/16 inch (24 mm) in diameter. Size 3 handles shall be 1-1/8 inch (29 mm) in diameter. The diameter tolerance shall be plus or minus 1/32 inch (0.8 mm).

3.4 Length. The handle length shall be specified by the contract or purchase order. Unless otherwise specified, the tolerance for the length specified shall be plus or minus 1/2 inch (1.3 cm). All handles shall be one piece and fabricated from a single piece of wood. The length shall be the overall length including any attached ends.

3.5 Finish. The handles, except for the tapered or ferrule end, shall be sanded smooth and coated with a durable, protective, clear, protective, and water resistant finish; such as varnish. The coating shall be continuous, smooth, uniform, and provide a glossy finish.

3.6 Special marking. Unless otherwise specified, each handle shall be permanently marked with the National Stock Number (NSN) assigned to that item along with the name or recognized trademark of the contractor. This marking shall begin within 2 feet (61 cm) of the attachment end of the handle. Permanent marking shall be by stamping with indelible ink, by a permanently attached water resistant label, or by burning the letters into the wood.

3.7 Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, providing they fall within the tolerances specified using conversion tables contained in the latest revision of FED-STD-376, and all other requirements are met. The metric dimensions provided in parenthesis are for information only, the inch-pound units shall apply.

3.8 Workmanship. The handles shall be clean, straight, free from raised grain, smoothly finished, free of mold, and free of any defects that may affect serviceability, durability, or appearance.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, contractors may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1 Inspection and tests. Samples for inspection and tests shall be selected in accordance with MIL-STD-105. A lot shall be all handles of one type, size and length offered for delivery at one time. A sample unit shall be one handle.

4.2.1 Inspection of end item. Examination of the end item shall be made for defects listed in table I. The inspection level shall be level I with Acceptable Quality Levels of 1.5 percent defective for major defects or 6.5 percent defective for total defects.

Table I. End item inspection, classification of defects

Defects	Classification Major	Minor	
Wood			
Not species specified.		X	
Any decay, mold, shake, heart center, or brashness.	X		
Not closed grained.		X	
Cross or spiral grain sloping more than specified (see 3.1).		X	
Crook, warp, or bow more than specified (see 3.2 and 4.2.1.1).		X	
Any knothole or loose knot.	X		
Any knot 1/8 inch (3.2 mm) or more in diameter.			
X			
Any season check, fracture or split - open or filled.	X		
Any wormhole, larva channel or pocket - open or filled.		X	
Any wane (lack of wood).	X		
Loose, raised or torn grain.	X		
Construction			
Handle not type specified.	X		
End opposite attachment end not rounded.		X	
Handle diameter not as specified.		X	
Length not as specified:			
- out of tolerance by less than 2.0 inches (5 cm).			X
- out of tolerance by 2.0 inches (5 cm) or more.		X	

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Table I, Continued	Classification Major	Minor
(Construction)		
Handle out of round.		X
Handle not of uniform diameter (see 3.2).		X
Handle composed of more than 1 piece.	X	
Type I		
No ferrule.	X	
Ferrule not threaded	X	
Ferrule not within specified dimensions.	X	
Threads not continuous or complete.	X	
Type II		
End not tapered.	X	
Taper out of round.		X
Tapered end not within specified dimensions.	X	
Finish		
Not coated.	X	
Coating not glossy.		X
Coating wet or tacky.	X	
Handle not smoothly finished.		X
Coating not continuous and uniform.		X
Dirt or foreign matter imbedded in coating.		X
Marking		
Incorrect, illegible, or missing.		X

4.2.1.1 Eccentricity. On a flat surface long enough for the entire handle to lie, roll the handle to determine the point that shows the greatest gap between the handle and the surface. This gap shall not exceed 3/8 inch (0.95 cm).

4.2.2 Testing of end item. Five (5) samples shall be selected for the following tests. Type II handles are tested for moisture content (4.2.2.2) only. Failure of one or more samples shall fail the lot.

4.2.2.1 Attachment of the ferrule (type I only). The samples shall be visually inspected for firm attachment. The handle shall be secured in a vertical position by the wood with the ferrule pointing down. A 10 pound weight shall be attached to the ferrule by a cord of sufficient length to allow the weight to free fall 12 inches (30.5 cm) without obstruction. The weight shall be allowed to drop once, falling a minimum of 12 inches (30.5 cm). The ferrule wood attachment shall be examined, a displacement of 1/8 inch (3.2 mm) or more shall fail the sample. As an alternative, securing of the ferrule and attachment of the weight to the wood is acceptable.

4.2.2.2 Moisture content. The finished handles shall be tested with a hand-held moisture meter in accordance with ASTM D 4444. The moisture shall not exceed 12%.

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4.3 Examination for preparation or delivery. A random sample of unit, intermediate and shipping container (as applicable) shall be selected from each lot and examined for conformance with the packaging, packing, and marking required in the contract or purchase order. Samples shall be selected in accordance with MIL-STD-105, inspection level S-2, AQL 6.5 defects per hundred units.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing and marking. The packaging, packing and marking shall be as specified on the contract or purchase order.

5.2 Special marking. Unless otherwise specified, each handle shall be marked as specified in paragraph 3.6.

6. NOTES

6.1 Intended use. Handles furnished this specification are intended as general use handles for use with brooms, brushes, squeegees, paint rollers, etc.

6.2 Ordering data. Purchasers should select the desired options offered in this document. Any procurement should specify the following:

(a) Title, number and date of this document.
(b) Type, size and length required (see 1.2).
(c) If special marking is not required (see 5.2).
(d) Packaging, packing and marking required.

6.3 Part numbering system. This part numbering system is intended for identification and cross-indexing of the item within the Federal cataloging system. Part numbers are not required to be placed on the product or container.

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- X: A - 4 feet (1.22 m); B - 4-1/2 feet (1.37 m); C - 5 feet (1.52 m); D - 8 feet (2.44 m)
- Y: 1 - Size 1 - 7/8 inch (22 mm) diameter; 2 - Size 2 - 15/16 inch (24 mm) diameter; 3 - Size 3 - 1-1/8 inch (29 mm) diameter
- Z: A - Type I - ACME threads; B - Type II - Tapered end

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6.4 National Stock Numbers (NSN's). The following NSN's reference this specification, this list may not be complete:

NSN	Part Number (see 6.3)
7920-00-982-6512	NNH104-A1A
7920-00-263-0328	NNH104-A2C
7920-00-177-5106	NNH104-B2B
7920-00-141-5452	NNH104-B3B
7920-00-263-0327	NNH104-B3D

MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an an official interest in this Federal Specification.

Preparing Activity
GSA-FSS

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